Futoshiki- Inequality puzzle

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Table of Content

1. Abstract
2. Introduction
3. Literature Review
4. Technology stacks
5. Image
6. References

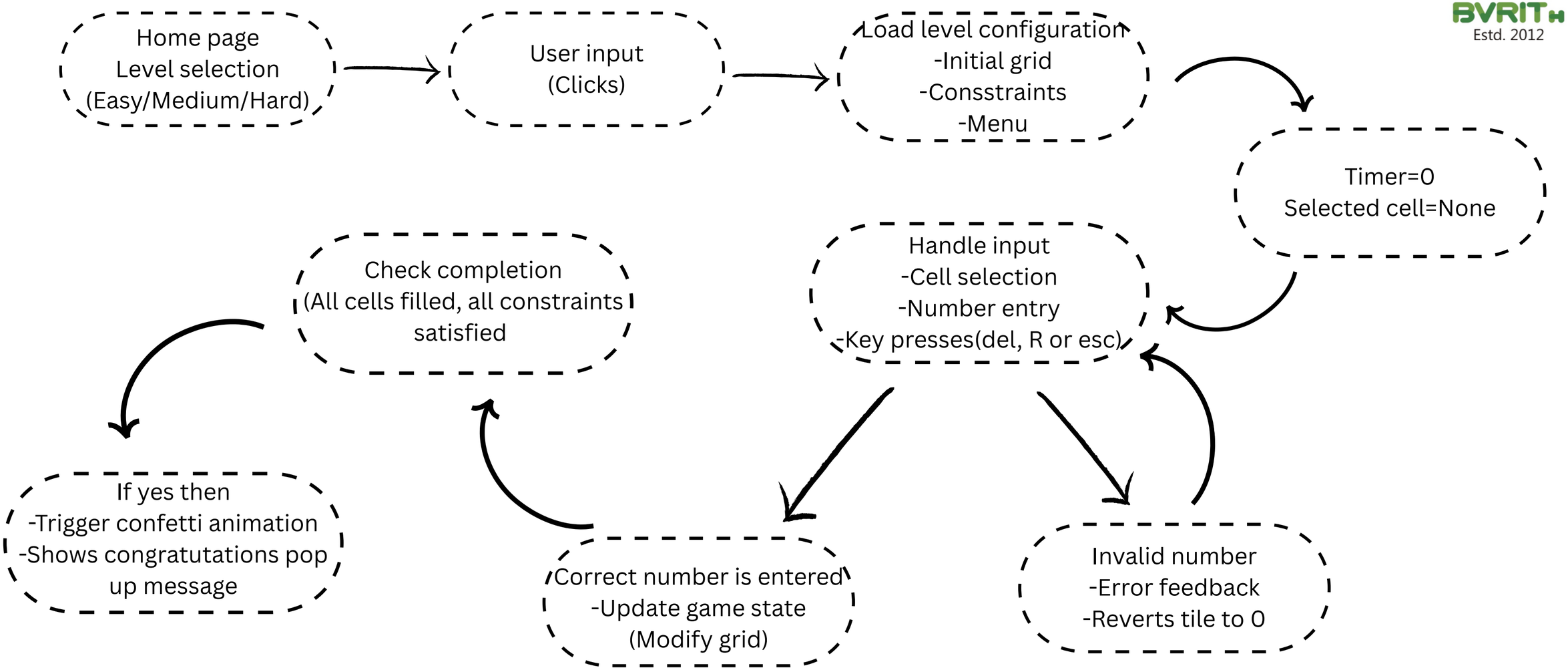
Abstract

This project describes the creation of an interactive digital Futoshiki game. The game features multiple levels of difficulty and puzzle generation, utilizing a 4x4 grid with specific numbers assigned to each row and column, adhering strictly to inequality restrictions, multiple difficulty tiers, and an automatic timer. This game is constructed using Python and offers a colourful, appealing, and creative user interface. The libraries used in this project are pygame, time, math and random. This project contributes to both educational entertainment and technological innovation. It also improves critical thinking and logical deduction. The outcome of this project is a completely interactive, operational Futoshiki game capable to check and verify riddles.

Introduction

The ‘Futoshiki’ is a logical game first introduced by Tamaki Seto in 2001. The main advantage of this game is it emphasizes cognitive development through interactive experiences. The comprehensive, user-friendly digital Futoshiki game aims to develop users' logical reasoning and deductive thinking skills. Its hallmarks include riddles created with adjustable difficulty levels with an easy-to-read user interface clearly portraying inequality limitations, made with Python. This project demonstrates how today's online technology can create complex and interactive logic games, not just simple brain training exercises. We designed and built a working game prototype from start to finish, focusing on making it a great experience for players.

Data flow diagram



Literature review

Reference/Tool	Focus area	Technique used	Limitations
Online Sudoku/Logic Puzzle Games (e.g., websudoku.com)	• User interface design • Basic input methods • Warnings on incorrect entries • Grid generation algorithms	• Graphical User Interface (GUI) frameworks • Constraint Satisfaction Problems (CSPs)	Primarily focused on Sudoku; less emphasis on inequality constraints unique to Futoshiki
Existing Open-Source Futoshiki Implementations	• Difficulty scaling • Puzzle generation with inequality constraints	• Custom constraint checking for inequalities	Limited in terms of interactive GUI, comprehensive features
Our Project	• Interactive Futoshiki gameplay • User-friendly GUI • Implementation for inequality	• Python (Pygame) • Constraint Satisfaction Problem (CSP) Formulation	Based on fixed dataset (pre-generated puzzles); it can be enhanced with advanced puzzle generation algorithms

Technology

Libraries/Packages	Where it is used	Real-life applications
pygame	• Creates display surface • Loads images and drawing shapes • Various fonts and font sizes • Checks for mouse clicks, key presses, etc. • Draws and animates confetti on screen	• 2D game development • Interactive simulations and displays • Graphical User Interfaces (GUIs)
time	• Starts automatic timer • Controls duration of confetti • Shows a visual feedback delay when invalid number is entered	• Scheduling and Tracking • Game development • Data analysis
random	• Generates random colours, sizes, speed and starting point of confetti	• Dice Rolls, Card Shuffles • Lottery and Gambling Simulations • Visual effects
math	• Simulates realistic, drift movement and visual rotation for the confetti particles	• Computer graphics • Navigation and GPS systems • Financial Modeling and Analytics

Images

Home page of Futoshiki game with customizable difficulty levels

Images

Game generated in “Easy” level futoshiki with a main menu along with automatic timer

Images

Grid tiles are highlighted in red as a warning when the user enters an incorrect number

Images

Confetti animation is generated when the game ends along with a pop-up message of the time taken to solve the futoshiki puzzle

References

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THANK YOU!